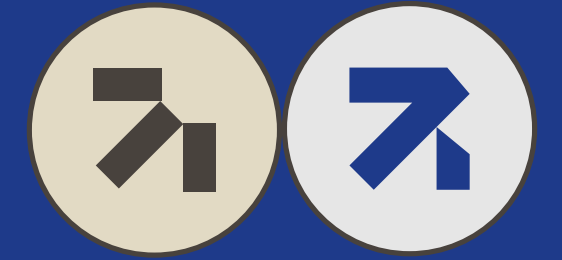
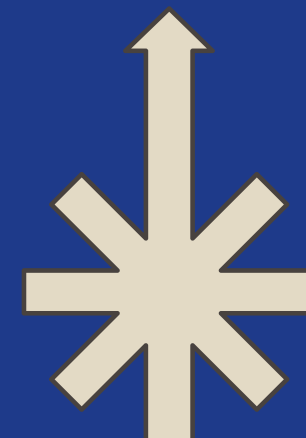


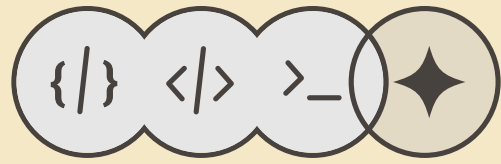
DATABASE MANAGEMENT SYSTEM



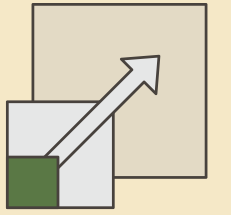
LABORATORY EQUIPMENT BOOKING & MAINTENANCE MANAGEMENT SYSTEM

SHAURYA DHINGRA - 25WU0102254
AIML WHALES

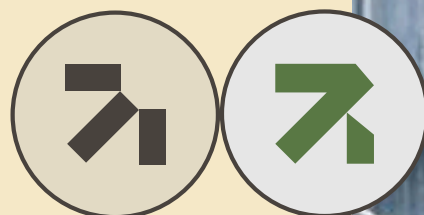




INTRODUCTION



- A database system for managing laboratory equipment, bookings, and maintenance.
- Allows students and faculty to book equipment based on availability..
- Tracks equipment maintenance, repairs, and maintenance schedules.
- Helps reduce booking conflicts and keeps equipment records organized.



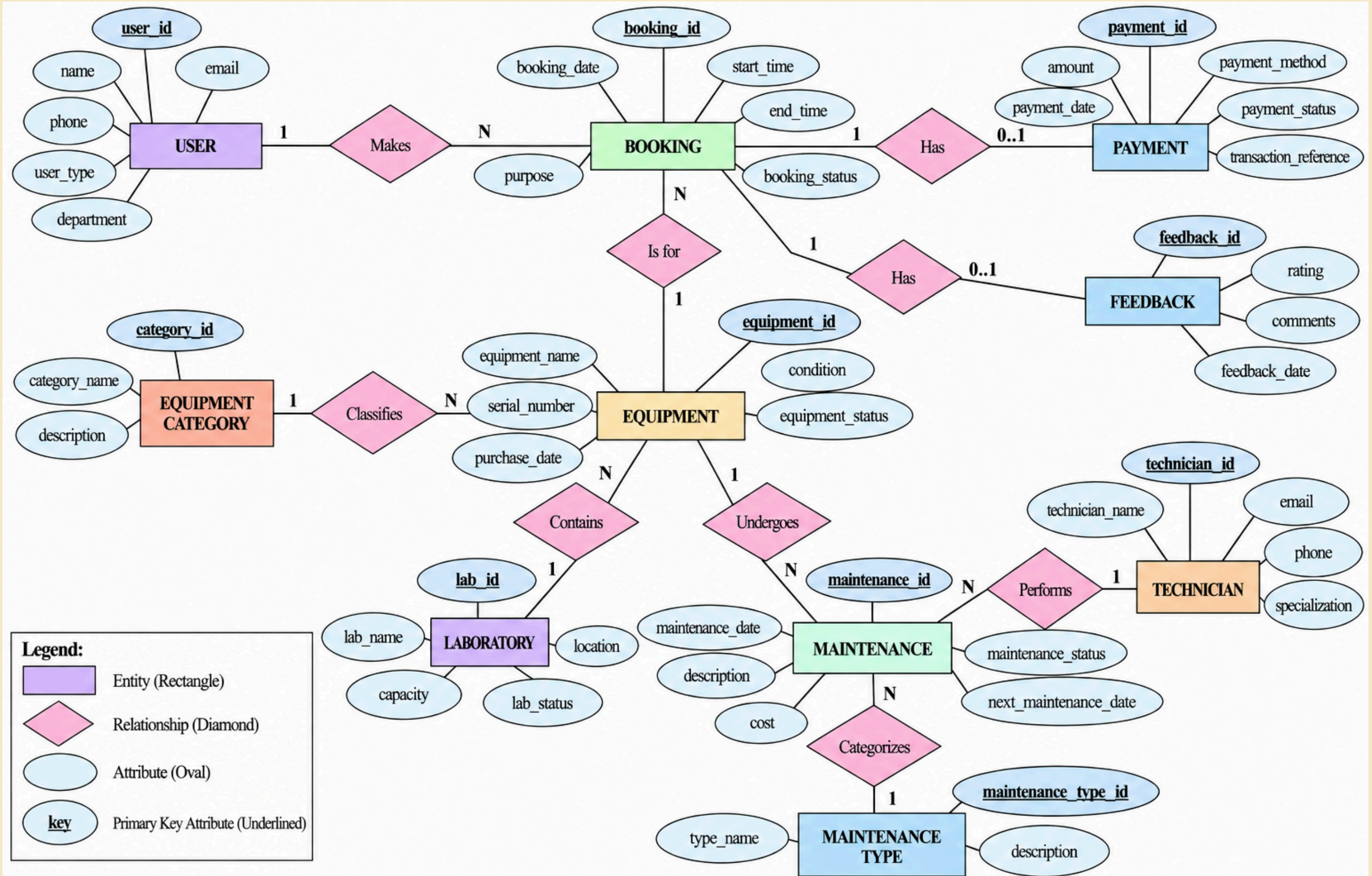
Goal

To efficiently manage laboratory equipment bookings, availability, maintenance, and user records through a centralized database system.

ER DIAGRAM

10 Principal Entities

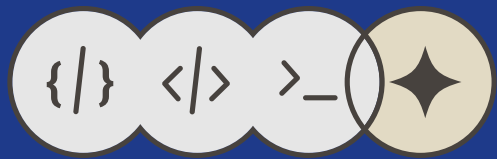
- USER
- LABORATORY
- EQUIPMENT_CATEGORY
- EQUIPMENT
- BOOKING
- TECHNICIAN
- MAINTENANCE_TYPE
- MAINTENANCE
- FEEDBACK
- PAYMENT



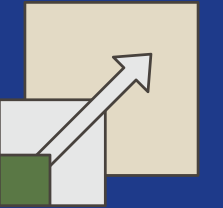
Initial Relational Schema

(Laboratory Equipment Booking and Maintenance System)

Table Name	Primary Key (PK)	Foreign Key (FK)
USER	user_id	—
LABORATORY	lab_id	—
EQUIPMENT_CATEGORY	category_id	—
EQUIPMENT	equipment_id	category_id, lab_id
BOOKING	booking_id	user_id, equipment_id
TECHNICIAN	technician_id	—
MAINTENANCE_TYPE	maintenance_type_id	—
MAINTENANCE	maintenance_id	equipment_id, technician_id, maintenance_type_id
FEEDBACK	feedback_id	booking_id
PAYMENT	payment_id	booking_id



DATABASE DESIGN



The system follows a relational database model with:

01 12 interconnected tables

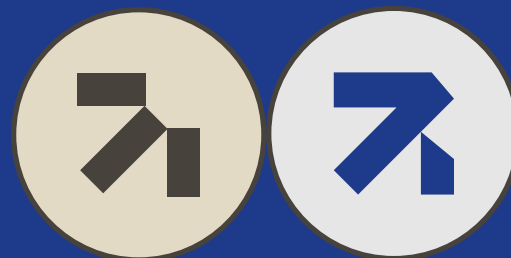
02 Primary keys for unique identification

03 Foreign keys for relationships

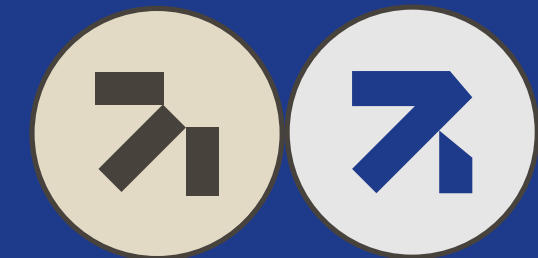
04 NOT NULL and UNIQUE constraints

05 CHECK constraints for valid values

06 DEFAULT values for common fields

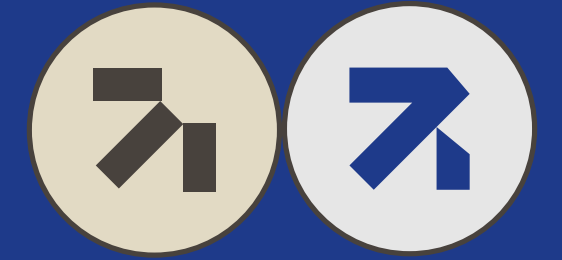


07 Triggers for automated business rules



Design Goal Maintain data integrity, consistency and minimize redundancy.

DATABASE MANAGEMENT SYSTEM



THANK
YOU

